

DANN Cost Model

OLAP Analysis Report

Ablation Study: Normalization Strategy & Domain Adaptation

1. Dataset Overview

The dataset contains 10,190 query samples across 4 domains spanning two database engines (MySQL and PostgreSQL), encoded across 79 features. The model architecture has 548,165 parameters and a feature dimensionality of 447.

1.1 Data Split

Train	Validation	Test
7,362 (~72.25%)	1,299 (~12.75%)	1,529 (~15.00%)

1.2 Domain Distribution & Class Weights

Domain	Engine	Count	Mean Cost	CE Weight
0	MySQL-32	1,898	17.19	0.1871
1	MySQL-64	102	5.62	3.4690
2	PostgreSQL-32	1,212	1.54	0.2930
3	PostgreSQL-64	6,978	2.74	0.0509

Note: MySQL domains (0, 1) have a mean cost of 16.60, versus 2.57 for PostgreSQL domains (2, 3) — a ~6.5x gap that motivates the choice of normalization strategy.

2. Normalization Strategies

2.1 Global Normalization

A single mean and standard deviation is computed over all rows. Every sample is normalized using the same global pair, and denormalization also uses the same global pair.

When it helps:

- Keeps a unified target scale across all domains.
- Better when all domains/engines are on a similar cost scale.
- Easier interpretation across the whole dataset.

Risk:

- If one engine/domain has much larger costs, it can dominate scaling and hurt smaller-scale groups.

2.2 Per-Engine Normalization

Separate mean and standard deviation are computed for each database engine. Each row is normalized using its own engine's statistics, and denormalization uses that row's engine stats.

When it helps:

- Handles strong engine-specific scale shifts (the MySQL vs. PostgreSQL cost gap).
- Makes each engine's target distribution closer to zero-mean, unit-standard-deviation.
- Often stabilizes multi-engine training.

3. Experiment Setup (A / B / C)

Label	Name	Description	alpha_target
A	Source Only	DANN with adversarial domain loss. Task loss uses source labels only. No labeled target samples.	0.0
B	SSDA 10%	Same DANN model with Semi-Supervised Domain Adaptation. Uses 10% labeled target subset in task loss.	0.3
C	Supervised	Same backbone/regression head, but no domain-adversarial objective. Plain supervised regression on all domains.	—

All three experiments use the same backbone and regression head. Seeds 42, 52, and 62 are used for robustness evaluation.

4. Evaluation Metrics

4.1 Normalized RMSE (NRMSE)

NRMSE expresses RMSE as a percentage of the average actual value, making it easy to state: "The model is off by X% on average." Formula: $NRMSE = RMSE / \text{mean}(\text{actual})$. If the average value is 100 and RMSE is 15, $NRMSE = 15\%$.

$$NRMSE = \frac{RMSE}{\bar{y}}$$

4.2 Mean Absolute Percentage Error (MAPE)

MAPE reports the average percentage error per data point. Formula: $MAPE = \text{mean}(|\text{actual} - \text{predicted}| / |\text{actual}|)$. It is intuitive for non-technical stakeholders, but can be unstable when actual values are close to zero.

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\text{Actual}_i - \text{Predicted}_i}{\text{Actual}_i} \right|$$

4.3 Spearman Rank Correlation (ρ)

A non-parametric measure of rank correlation between predicted and actual values. A value closer to 1.0 indicates better rank ordering of query costs.

5. Per-Run Results

All 18 runs (3 experiments $\times$ 2 normalization modes $\times$ 3 seeds). Sorted by normalization mode, seed, and experiment.

Exp	Norm. Mode	Seed	RMSE	NRMSE %	MAPE %	Spearman ρ	Domain Acc
B	global	42	6.52	10.51%	285.16%	0.488	0.305
C	global	42	6.54	10.53%	199.82%	0.768	0.010
A	global	42	7.29	11.74%	121.53%	0.705	0.695
C	global	52	6.46	9.63%	281.86%	0.672	0.010
B	global	52	6.49	9.68%	286.93%	0.680	0.548
A	global	52	7.20	10.75%	129.92%	0.792	0.483
B	global	62	124.07	2.55%	166.45%	0.725	0.674
C	global	62	124.09	2.55%	60.08%	0.897	0.379
A	global	62	124.44	2.56%	72.90%	0.699	0.299
B	per_engine	42	6.96	11.21%	135.13%	0.817	0.605
C	per_engine	42	7.07	11.39%	119.43%	0.876	0.239
A	per_engine	42	83.99	135.29%	1158.69%	0.813	0.664
C	per_engine	52	7.83	11.68%	120.26%	0.766	0.186
B	per_engine	52	10.19	15.20%	160.33%	0.837	0.795
A	per_engine	52	122.06	182.11%	1727.04%	0.810	0.777
A	per_engine	62	123.94	2.55%	237.41%	0.788	0.531
B	per_engine	62	124.04	2.55%	315.67%	0.524	0.307
C	per_engine	62	124.22	2.56%	60.00%	0.886	0.179

Note: Seed 62 is an anomalous run — all experiments produce RMSE $\sim$ 124 regardless of normalization mode, significantly inflating cross-seed averages and standard deviations.

6. Robust Summary (Mean ± Std Across Seeds)

Ranks are assigned based on mean RMSE across the three seeds for each normalization mode. Green rows indicate rank-1 winners; red rows indicate rank-3 (worst).

Ran k	Mode	Experiment	RMSE ± std	NRMSE %	MAPE %	Spearman ρ	Domain Acc
1	global	B — DANN SSDA 10%	45.69 ± 67.87	7.58 ± 4.37%	246.18%	0.631	0.509
2	global	C — No DANN Supervised	45.70 ± 67.89	7.57 ± 4.37%	180.59%	0.779	0.133
3	global	A — DANN Source Only	46.31 ± 67.66	8.35 ± 5.04%	108.12%	0.732	0.492
1	per_engi ne	C — No DANN Supervised	46.37 ± 67.42	8.54 ± 5.19%	99.90%	0.843	0.201
2	per_engi ne	B — DANN SSDA 10%	47.06 ± 66.69	9.65 ± 6.47%	203.71%	0.726	0.569
3	per_engi ne	A — DANN Source Only	109.99 ± 22.54	106.65 ± 93.14%	1041.05%	0.804	0.658

7. Per-Run Winners (Lowest RMSE per Mode & Seed)

Normalization Mode	Seed	Winning Experiment	RMSE
global	42	B — DANN SSDA 10%	6.52
global	52	C — No DANN Supervised	6.46
global	62	B — DANN SSDA 10%	124.07 ⚠
per_engine	42	B — DANN SSDA 10%	6.96
per_engine	52	C — No DANN Supervised	7.83
per_engine	62	A — DANN Source Only	123.94 ⚠

Experiment A	Experiment B	Experiment C
1 win	3 wins	2 wins

8. Conclusion

The ablation study reveals that model performance is strongly influenced by both normalization strategy and data distribution variability across random seeds.

Global Normalization	Per-Engine Normalization
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Rank 1: B (DANN SSDA 10%) RMSE 45.69 NRMSE 7.58% MAPE 246% B and C are near-tied on RMSE/NRMSE. C leads on Spearman rho and MAPE. Semi-supervised adaptation provides a slight but situational benefit.	Rank 1: C (No DANN Supervised) RMSE 46.37 NRMSE 8.54% MAPE 99.9% C wins clearly. A collapses badly (NRMSE 107%, MAPE 1041%). When domain discrepancies are minimised by normalization, plain supervised learning is sufficient and preferable.
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Key Findings

- Normalization strategy plays a critical role in cross-domain performance.
- Semi-supervised domain adaptation (B) offers limited but situational benefits over plain supervised (C).
- Source-only DANN (A) performs worst overall and exhibits severe instability under per-engine normalization.
- Higher domain classification accuracy does not correlate with better predictive performance, highlighting a limitation of adversarial domain adaptation.
- Model evaluation must consider multiple random seeds due to high sensitivity to data splits (seed 62 anomaly).